

RL Motion Tracking Implementation Manual

Based on the BeyondMimic / Perceptive Humanoid Parkour Pipeline

From Motion-Matching Datasets to Trained Expert Policies

Reference: Wu et al., *Perceptive Humanoid Parkour* (2025)
Liao et al., *BeyondMimic* (2025)

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Abstract

This manual provides a comprehensive, step-by-step guide to implementing the Reinforcement Learning (RL) motion tracking pipeline used in the Perceptive Humanoid Parkour (PHP) framework. Assuming the reader has already generated motion-matching kinematic reference trajectories, this document covers the principles of the BeyondMimic-style motion tracking algorithm, the Markov Decision Process (MDP) formulation, reward design, observation and action spaces, training procedures with Proximal Policy Optimization (PPO), domain randomization strategies, adaptive sampling, and practical implementation details using Isaac Lab. The manual is designed as a hands-on reference for researchers and engineers who wish to train motion-tracking expert policies for humanoid robots.

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强化学习运动跟踪实现手册

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摘要

本手册提供了实现感知仿人跑酷（PHP）框架中基于强化学习（RL）运动跟踪流程的全面、逐步指南。假设读者已生成运动匹配的运动学参考轨迹，本文档涵盖了 BeyondMimic-style 运动跟踪算法的原理、马尔可夫决策过程（MDP）公式、奖励设计、观测与动作空间、基于近端策略优化（PPO）的训练流程、域随机化策略、自适应采样，以及使用 Isaac Lab 的实际实现细节。该手册旨在为希望训练仿人机器人运动跟踪专家策略的研究人员和工程师提供实践参考。

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1 Introduction and Motivation

The RL Motion Tracking stage is the second major component in the Perceptive Humanoid Parkour (PHP) pipeline, sitting between kinematic skill composition (motion matching) and multi-skill student distillation. Its purpose is to train physics-based controllers—called *expert policies*—that can faithfully reproduce kinematic reference motions on a simulated humanoid robot subject to full rigid-body dynamics.

The core methodology follows the BeyondMimic framework, which itself builds upon the DeepMimic paradigm. The key insight is:

Core Principle

Given a kinematic reference trajectory $\{q_t^{\text{ref}}\}_{t=0}^T$, train a policy $\pi_\theta(a_t | o_t)$ via reinforcement learning so that the physically simulated robot tracks the reference motion as closely as possible, while remaining robust to perturbations through domain randomization.

The trained expert policies serve two purposes in the PHP pipeline:

1. They produce high-quality, physically feasible motion execution for each individual skill.
2. They serve as teachers for the downstream DAgger+RL distillation stage, providing expert actions that a single multi-skill student policy can imitate.

1.1 Scope of This Manual

This manual assumes:

- You have successfully generated motion-matching reference trajectories (the kinematic parkour trajectories from Section III-B of the PHP paper).
- Reference motions have been retargeted to the target robot (e.g., Unitree G1 with 29 DoFs) using a tool such as OmniRetarget.
- You have access to a GPU-accelerated physics simulator (Isaac Sim / Isaac Lab) with parallel environment support.

2 Algorithm Principles

2.1 The DeepMimic Paradigm

DeepMimic (Peng et al., 2018) introduced the idea of using human motion capture data as dense reference signals for RL-based character control. The agent receives a reward proportional to how closely its current physical state matches the reference motion at each timestep. This avoids the need for hand-crafted task-specific reward functions and naturally produces human-like motions.

The reward is typically formulated as a product (or weighted sum) of Gaussian-shaped tracking scores for different body quantities—positions, orientations, velocities, and angular velocities of tracked bodies.

2.2 BeyondMimic Enhancements

BeyondMimic extends DeepMimic with several critical improvements that make the pipeline scalable and hardware-deployable:

1. **Unified MDP formulation.** A single set of hyperparameters works across all motions—from walking to cartwheels—without per-motion tuning.

1 引言与动机

RL 运动跟踪阶段是感知型人形跑酷（PHP）流水线中的第二个主要组件，位于运动学技能组合（运动匹配）与多技能学生蒸馏之间。其目标是训练基于物理的控制器——称为专家策略——使其能够在模拟人形机器人上忠实再现运动学参考运动，并受完整刚体动力学的约束。

核心方法遵循 BeyondMimic 框架，而该框架本身建立在 DeepMimic 范式之上。关键见解是：

核心原理

给定运动学参考轨迹 $\{q_t^{\text{ref}}\}_{t=0}^T$ ，通过强化学习训练策略 $\pi_\theta(a_t | o_t)$ ，使物理仿真机器人尽可能精确地跟踪参考运动，同时通过域随机化对扰动保持鲁棒性。

在 PHP 流水线中，训练好的专家策略服务于两个目的：

1. 它们针对每个单独技能生成高质量、物理上可行的运动执行。
2. 它们作为下游 DAgger+RL 蒸馏阶段的教师，提供专家动作以供单一多技能学生策略模仿。

1.1 本手册的范围

本手册假定：

已成功生成运动匹配参考轨迹（即 PHP 论文第 III-B 节中的运动学跑酷轨迹）。

使用 OmniRetarget 等工具，参考运动已被重定向到目标机器人（例如具有 29 个自由度的 Unitree G1）。

你可以使用一个支持并行环境的 GPU 加速物理模拟器（Isaac Sim / Isaac Lab）。

2 算法原理

2.1 DeepMimic 范式

DeepMimic (Peng et al., 2018) 提出了利用人体运动捕捉数据作为基于强化学习的角色控制的密集参考信号。智能体在每个时间步获得的奖励与其当前物理状态匹配参考运动的程度成正比。这避免了手工设计特定任务的奖励函数，并自然生成类人运动。

该奖励通常表示为跟踪身体不同物理量（位置、方向、速度及角速度）的高斯型跟踪得分的乘积（或加权和）。

2.2 BeyondMimic 增强

BeyondMimic 对 DeepMimic 进行了多项关键改进，使其流程可扩展且可直接部署到硬件上：

1. **统一 MDP 公式：**一组超参数即可适用于所有运动——从步行到侧手翻——无需针对每种运动进行调参。

2. **Anchor-based tracking.** The reference motion is anchored to a selected body (typically the pelvis/torso), allowing global drift while preserving motion style. This separates *style tracking* from *global position tracking*.
3. **Adaptive sampling.** Training episodes are preferentially initialized at difficult segments of the reference motion, accelerating learning on challenging contact-rich phases.
4. **Careful sim-to-real design.** Joint impedance parameters are set heuristically for compliance, and lightweight domain randomization is applied to bridge the sim-to-real gap without degrading motion quality.

2.3 Key Differences in PHP’s Usage

The PHP paper adopts BeyondMimic’s motion tracking formulation but adds:

- **Global tracking with privileged observations.** The expert receives the pelvis global position and velocity as privileged information, enabling it to learn recovery behaviors when drift accumulates relative to terrain-coupled references.
- **Height scan observations.** A $0.7m \times 0.7m$ terrain height map is provided to the expert so it can adapt to terrain randomizations around obstacles.
- **Per-skill expert training.** Rather than training one policy for all motions, PHP trains separate expert policies for each parkour skill, which are later distilled.
- **Adaptive sampling for difficult skills.** For challenging maneuvers like high-wall climbing, adaptive sampling is essential for convergence.

3 MDP Formulation

The motion tracking problem is formulated as a Markov Decision Process (MDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$.

3.1 State / Observation Space

The expert policy’s observation vector o_t consists of:

Table 1: Expert policy observation components.

Component	Description	Dim.
Reference joint positions	Target joint angles θ_t^{ref}	n_j
Reference joint velocities	Target joint angular velocities $\dot{\theta}_t^{\text{ref}}$	n_j
Reference pelvis pose error	Position & orientation error to reference pelvis	6
Pelvis linear velocity	Current pelvis linear velocity	3
Pelvis angular velocity	Current pelvis angular velocity	3
Joint positions	Current joint angles θ_t	n_j
Joint velocities	Current joint angular velocities $\dot{\theta}_t$	n_j
Previous action	Action from the last timestep a_{t-1}	n_j
<i>Privileged observations (expert only):</i>		
Pelvis global position	Global (x, y, z) of pelvis	3
Pelvis global velocity	Global $(\dot{x}, \dot{y}, \dot{z})$ of pelvis	3
Height scan	$0.7m \times 0.7m$ terrain heightmap	varies

For the Unitree G1, $n_j = 29$ degrees of freedom.

2. 基于锚点的跟踪。参考运动被锚定至选定的身体部位（通常为骨盆/躯干），允许全局漂移的同时保持运动风格。这便将风格跟踪与全局位置跟踪分离开来。
3. 自适应采样。训练回合优先在参考运动的困难片段处初始化，从而加速在挑战性接触丰富阶段的学习。
4. 精细的仿真到现实设计。关节阻抗参数通过启发式方法设置为柔顺性，并应用轻量级域随机化以弥合仿真到现实之间的差距，同时不降低运动质量。

2.3 PHP 使用的关键差异

PHP 论文采用了 BeyondMimic 的运动跟踪公式，但添加了：

- 全局跟踪与特权观测。专家接收骨盆全局位置与速度作为特权信息，使其能够学习在漂移相对于地形耦合参考累积时的恢复行为。
 - 高度扫描观测。向专家提供一张 $0.7m \times 0.7m$ 的地形高度图，使其能够适应障碍物周围的地形随机化。
- 逐技能专家训练。与训练一个策略处理所有动作不同，PHP 为每个跑酷技能分别训练专家策略，这些策略在后期进行知识蒸馏。
- 针对困难技能的自适应采样。对于高墙攀爬等高难度动作，自适应采样是保证收敛的关键。

3 MDP 形式化

运动跟踪问题被形式化为一个马尔可夫决策过程（MDP） $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$ 。

3.1 状态/观测空间

专家策略的观测向量 o_t 由以下部分组成：

表 1: 专家策略观测组成部分

Component	Description	Dim.
Reference joint positions	Target joint angles θ_t^{ref}	n_j
Reference joint velocities	Target joint angular velocities $\dot{\theta}_t^{\text{ref}}$	n_j
Reference pelvis pose error	Position & orientation error to reference pelvis	6
Pelvis linear velocity	Current pelvis linear velocity	3
Pelvis angular velocity	Current pelvis angular velocity	3
Joint positions	Current joint angles θ_t	n_j
Joint velocities	Current joint angular velocities $\dot{\theta}_t$	n_j
Previous action	Action from the last timestep a_{t-1}	n_j
<i>Privileged observations (expert only):</i>		
Pelvis global position	Global (x, y, z) of pelvis	3
Pelvis global velocity	Global $(\dot{x}, \dot{y}, \dot{z})$ of pelvis	3
Height scan	$0.7m \times 0.7m$ terrain heightmap	varies

对于 Unitree G1，自由度 $n_j = 29$ 。

3.2 Action Space

Actions $a_t \in \mathbb{R}^{n_j}$ are **normalized joint position (PD) targets**. They serve as setpoints for a proportional-derivative (PD) controller that converts position targets to joint torques:

$$\tau_j = k_p^j(\hat{\theta}_j - \theta_j) - k_d^j\dot{\theta}_j \quad (1)$$

where $\hat{\theta}_j = \theta_j^{\text{default}} + a_j \cdot s$ is the target position, s is the **action scale** (set to 1 in PHP for all experts), k_p^j and k_d^j are the PD gains for joint j .

Implementation Note

In PHP, the action scale is set to 1 for all experts, unlike the heuristic scaling used in the original BeyondMimic. This is because parkour skills require challenging RL exploration, and a uniform scale simplifies the setup. The PD gains (k_p , k_d) are set heuristically to promote compliance for hardware deployment.

3.3 Reward Function

The reward follows the DeepMimic-style formulation using Gaussian-shaped tracking scores. Each reward term measures the discrepancy between the simulated robot state and the reference motion, mapped through an exponential kernel:

$$r = \prod_k \exp\left(-\frac{\text{error}_k^2}{\sigma_k^2}\right) \quad \text{or equivalently} \quad r = \sum_k w_k \exp\left(-\frac{\text{error}_k^2}{\sigma_k^2}\right) \quad (2)$$

The specific reward terms (from the PHP paper, Table IV) are:

Term	Formula	Weight
Body Position	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ p_b^{\text{des}} - p_b\ ^2 / 0.3^2\right)$	1.0
Body Orientation	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ \log(R_b^{\text{des}} R_b^\top)\ ^2 / 0.4^2\right)$	1.0
Body Linear Vel.	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ v_b^{\text{des}} - v_b\ ^2 / 1.0^2\right)$	1.0
Body Angular Vel.	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ \omega_b^{\text{des}} - \omega_b\ ^2 / 3.14^2\right)$	1.0
Anchor Position	$\exp(-\ p_{\text{anc}}^{\text{des}} - p_{\text{anc}}\ ^2 / 0.3^2)$	1.0
Anchor Orientation	$\exp(-\ \log(R_{\text{anc}}^{\text{des}} R_{\text{anc}}^\top)\ ^2 / 0.4^2)$	1.0
<i>Regularization (penalties):</i>		
Action smoothness	$\ a_t - a_{t-1}\ ^2$	-0.1
Joint position limit	$\sum_j [\max(l_j - \theta_j, 0) + \max(\theta_j - u_j, 0)]$	-10.0
Undesired contacts	$\sum_{b \notin B_{\text{ee}}} \mathbf{1}[\ f_b^{\text{self}}\ > 1\text{N}]$	-0.5

Here, B is the set of tracked bodies, p_b , R_b , v_b , ω_b are the position, rotation, linear velocity, and angular velocity of body b , and “anc” denotes the anchor body (pelvis). The σ values (0.3, 0.4, 1.0, 3.14) control the sensitivity of each tracking term.

3.4 Episode Initialization and Termination

Initialization. Episodes are initialized by sampling a random frame t_0 from the reference trajectory and resetting the simulated robot to the corresponding reference pose $q_{t_0}^{\text{ref}}$. With adaptive sampling, frames where the policy historically fails are sampled more frequently.

3.2 动作空间

动作 $a_t \in \mathbb{R}^{n_j}$ 是归一化的关节位置 (PD) 目标, 作为比例-微分 (PD) 控制器的设定点, 将位置目标转换为关节力矩:

$$\tau_j = k_p^j(\hat{\theta}_j - \theta_j) - k_d^j\dot{\theta}_j \quad (1)$$

其中 $\hat{\theta}_j = \theta_j^{\text{default}} + a_j \cdot s$ 是目标位置, s 是动作尺度 (在 PHP 中所有专家均设为 1), k_p^j 和 k_d^j 是关节 j 的 PD 增益。

实现说明

在 PHP 中, 所有专家的动作缩放 (action scale) 均设为 1, 这与原始 BeyondMimic 中使用的启发式缩放不同。这是因为跑酷技能需要具有挑战性的强化学习探索, 而统一的缩放简化了设置。PD 增益 (k_p , k_d) 采用启发式设定, 以提升硬件部署时的顺应性。

3.3 奖励函数

奖励函数采用 DeepMimic 风格的框架, 使用高斯形状跟踪得分。每个奖励项衡量仿真机器人状态与参考运动之间的差异, 并通过指数核进行映射:

$$r = \prod_k \exp\left(-\frac{\text{error}_k^2}{\sigma_k^2}\right) \quad \text{or equivalently} \quad r = \sum_k w_k \exp\left(-\frac{\text{error}_k^2}{\sigma_k^2}\right) \quad (2)$$

具体的奖励项 (来自 PHP 论文, 表 IV) 为:

表 2: 奖励函数——高斯形状跟踪分数。

Term	Formula	Weight
Body Position	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ p_b^{\text{des}} - p_b\ ^2 / 0.3^2\right)$	1.0
Body Orientation	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ \log(R_b^{\text{des}} R_b^\top)\ ^2 / 0.4^2\right)$	1.0
Body Linear Vel.	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ v_b^{\text{des}} - v_b\ ^2 / 1.0^2\right)$	1.0
Body Angular Vel.	$\exp\left(-\frac{1}{ B } \sum_{b \in B} \ \omega_b^{\text{des}} - \omega_b\ ^2 / 3.14^2\right)$	1.0
Anchor Position	$\exp(-\ p_{\text{anc}}^{\text{des}} - p_{\text{anc}}\ ^2 / 0.3^2)$	1.0
Anchor Orientation	$\exp(-\ \log(R_{\text{anc}}^{\text{des}} R_{\text{anc}}^\top)\ ^2 / 0.4^2)$	1.0
<i>Regularization (penalties):</i>		
Action smoothness	$\ a_t - a_{t-1}\ ^2$	-0.1
Joint position limit	$\sum_j [\max(l_j - \theta_j, 0) + \max(\theta_j - u_j, 0)]$	-10.0
Undesired contacts	$\sum_{b \notin B_{\text{ee}}} \mathbf{1}[\ f_b^{\text{self}}\ > 1\text{N}]$	-0.5

这里, B 是被跟踪刚体的集合, p_b 、 R_b 、 v_b 、 ω_b 分别为刚体 b 的位置、旋转、线速度和角速度, “anc” 表示锚点刚体 (骨盆)。 σ 值 (0.3、0.4、1.0、3.14) 控制每个跟踪项的灵敏度。

3.4 回合初始化与终止

初始化。回合的初始化方式为: 从参考轨迹中随机采样一帧 t_0 , 并将仿真机器人重置到对应的参考位姿 $q_{t_0}^{\text{ref}}$ 。通过自适应采样, 策略历史上失败的帧会以更高的频率被采样。

Early Termination. A tracking-based early termination is triggered when the anchor (pelvis) deviates from the reference by more than a threshold d_{term} :

$$\|p_{\text{anc}} - p_{\text{anc}}^{\text{ref}}\| > d_{\text{term}} \quad (3)$$

In the expert training, $d_{\text{term}} = 0.5\text{m}$. (This is relaxed to 1.0m during student distillation to accommodate mirrored executions.)

Timeout. Episodes also terminate upon reaching the end of the reference clip or a maximum episode length.

4 Training Algorithm: PPO with Motion Tracking

4.1 Proximal Policy Optimization (PPO)

The expert policies are trained using PPO, an on-policy actor-critic algorithm. PPO optimizes a clipped surrogate objective:

$$L^{\text{PPO}}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \quad (4)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|o_t)}{\pi_{\theta_{\text{old}}}(a_t|o_t)}$ is the probability ratio, $\hat{A}_t$ is the estimated advantage, and $\epsilon = 0.2$ is the clipping parameter.

Advantages are estimated using Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(o_{t+1}) - V(o_t) \quad (5)$$

with $\gamma = 0.99$ and $\lambda_{\text{GAE}} = 0.95$.

4.2 Asymmetric Actor-Critic

The system uses an **asymmetric actor-critic** architecture:

- The **actor** (policy) receives only the standard observations available on hardware.
- The **critic** (value function) receives privileged information including global position, global velocity, and terrain heightmaps.

This allows the critic to provide better value estimates during training, while the actor learns to operate with limited information.

4.3 Network Architecture

Table 3: Network architecture for expert policies.

Component	Specification
Actor MLP hidden dims	[512, 256, 128]
Critic MLP hidden dims	[512, 256, 128]
Activation function	ELU
Initial noise std	1.0

提前终止。当锚点（骨盆）偏离参考位置超过阈值 d_{term} 时，触发基于跟踪的提前终止：

$$\|p_{\text{anc}} - p_{\text{anc}}^{\text{ref}}\| > d_{\text{term}} \quad (3)$$

在专家训练中, $d_{\text{term}} = 0.5\text{m}$ 。(在学生蒸馏过程中, 该值放宽至 1.0m, 以适应镜像执行。)

超时。当到达参考片段的末尾或达到最大回合长度时, 回合也会终止。

4 训练算法：结合运动跟踪的 PPO

4.1 近端策略优化 (PPO)

专家策略使用 PPO (一种在策略演员-评论家算法) 进行训练。PPO 优化一个裁剪的替代目标：

$$L^{\text{PPO}}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \quad (4)$$

其中 $r_t(\theta) = \frac{\pi_\theta(a_t|o_t)}{\pi_{\theta_{\text{old}}}(a_t|o_t)}$ 为概率比, $\hat{A}_t$ 为估计优势, $\epsilon = 0.2$ 为裁剪参数。

优势采用广义优势估计 (GAE) 计算：

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(o_{t+1}) - V(o_t) \quad (5)$$

其中 $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$ 。

4.2 非对称演员-评论家

该系统采用非对称演员-评论家架构：

- 动作网络（策略）仅接收硬件上可用的标准观测数据。
- 评论家（价值函数）接收特权信息，包括全局位置、全局速度和地形高度图。

这使得评论家在训练过程中能够提供更优的价值估计，而演员则学会在有限信息下运行。

4.3 网络架构

表 3: 专家策略的网络架构。

Component	Specification
Actor MLP hidden dims	[512, 256, 128]
Critic MLP hidden dims	[512, 256, 128]
Activation function	ELU
Initial noise std	1.0

Table 4: PPO training hyperparameters for expert motion tracking.

Hyperparameter	Value
Steps per environment	24
Max iterations	20,000
Learning rate	1×10^{-3}
Schedule	Adaptive
Clip parameter ϵ	0.2
Entropy coefficient	0.005
Discount factor γ	0.99
GAE λ	0.95
Desired KL	0.01
Learning epochs per update	5
Mini-batches	4
Max gradient norm	1.0
Parallel environments	16,384

4.4 Training Hyperparameters

5 Adaptive Sampling

Adaptive sampling is essential for learning difficult skills such as high-wall climbing. The idea is to preferentially initialize training episodes at reference frames where the policy has historically failed, rather than sampling uniformly across all frames.

Algorithm 1 Adaptive Sampling for Expert Training

- 1: **Initialize:** failure counts $F[i] = 0$ for all reference frames i
 - 2: **for** each training iteration **do**
 - 3: Compute sampling probability: $P(i) \propto F[i] + \epsilon_{\text{base}}$
 - 4: Sample start frames $\{t_0^{(k)}\}_{k=1}^{N_{\text{env}}}$ according to P
 - 5: Roll out episodes from sampled start frames
 - 6: Update $F[i]$ based on which frames led to early termination
 - 7: Update policy with PPO
 - 8: **end for**
-

Without adaptive sampling, the policy may converge on easy segments (e.g., walking) but never learn the critical contact-rich phases (e.g., the pull-up phase of wall climbing) because those frames are rarely sampled.

6 Domain Randomization

Domain randomization is applied during expert training to improve sim-to-real robustness. The randomizations are intentionally lightweight to avoid degrading motion quality.

7 Implementation Guide

7.1 Prerequisites

1. **Isaac Lab v2.1.0+** installed with IsaacSim 4.5.0 backend.
2. **Python 3.10** with conda environment recommended.

表 4: PPO 用于专家运动的训练超参数跟踪。

Hyperparameter	Value
Steps per environment	24
Max iterations	20,000
Learning rate	1×10^{-3}
Schedule	Adaptive
Clip parameter ϵ	0.2
Entropy coefficient	0.005
Discount factor γ	0.99
GAE λ	0.95
Desired KL	0.01
Learning epochs per update	5
Mini-batches	4
Max gradient norm	1.0
Parallel environments	16,384

4.4 训练超参数

5 自适应采样

自适应采样对于学习高墙攀爬等高难度技能至关重要。其思想是优先在策略历史上失败的参考帧处初始化训练回合，而非在所有帧上均匀采样。

Algorithm 1 Adaptive Sampling for Expert Training

- 1: **Initialize:** failure counts $F[i] = 0$ for all reference frames i
 - 2: **for** each training iteration **do**
 - 3: Compute sampling probability: $P(i) \propto F[i] + \epsilon_{\text{base}}$
 - 4: Sample start frames $\{t_0^{(k)}\}_{k=1}^{N_{\text{env}}}$ according to P
 - 5: Roll out episodes from sampled start frames
 - 6: Update $F[i]$ based on which frames led to early termination
 - 7: Update policy with PPO
 - 8: **end for**
-

如果没有自适应采样，策略可能会在简单片段（如行走）上收敛，但永远不会学习关键的高接触力阶段（如爬墙中的引体向上阶段），因为那些帧很少被采样。

6 域随机化

在专家训练过程中应用域随机化，以提升仿真到现实的鲁棒性。随机化被有意设计为轻量级，从而避免运动质量下降。

7 实现指南

7.1 先决条件

1. Isaac Lab v2.1.0 + 安装于 IsaacSim 4.5.0 后端。
2. 推荐使用 Python 3.10 和 conda 环境。

Table 5: Domain randomization parameters.

Parameter	Sampling Distribution
<i>Physical parameters</i>	
Static friction μ_{static}	$\mathcal{U}[0.4, 1.3]$
Dynamic friction μ_{dynamic}	$\mathcal{U}[0.4, 1.1]$
Restitution e_{rest}	$\mathcal{U}[0, 0.5]$
Default joint positions (non-ankle)	$\Delta\theta_j^0 \sim \mathcal{U}[-0.01, 0.01]$ rad
Default ankle joint positions	$\Delta\theta_j^0 \sim \mathcal{U}[-0.03, 0.03]$ rad
Torso COM offset	$\Delta x \sim \mathcal{U}[-0.025, 0.025]$ m
	$\Delta y, \Delta z \sim \mathcal{U}[-0.05, 0.05]$ m
<i>Root velocity perturbations</i>	
Root linear velocity	$v_x, v_y \sim \mathcal{U}[-0.1, 0.1]$ m/s
	$v_z \sim \mathcal{U}[-0.05, 0.05]$ m/s
Push duration	$\Delta t \sim \mathcal{U}[1.0, 3.0]$ s
Root angular velocity	$\omega_x, \omega_y, \omega_z \sim \mathcal{U}[-0.1, 0.1]$ rad/s

3. BeyondMimic codebase: https://github.com/HybridRobotics/whole_body_tracking
4. RSL-RL PPO implementation (bundled with Isaac Lab).
5. GPU with sufficient VRAM for 16K+ parallel environments (NVIDIA A100 or similar).

7.2 Step 1: Prepare Reference Motion Data

Your motion-matching output should be a sequence of generalized coordinates per timestep. Convert to the required format containing maximum coordinates (body pose, velocity, acceleration) via forward kinematics:

Listing 1: Convert reference motion to training format.

```

1 # Convert CSV reference to NPZ with full kinematic info
2 python scripts/csv_to_npz.py \
3     --input_file parkour_skill_climb_76cm.csv \
4     --input_fps 30 \
5     --output_name climb_76cm \
6     --headless

```

The reference motion file should contain per-frame:

- Root translation $p_t \in \mathbb{R}^3$
- Root quaternion $r_t \in \mathbb{R}^4$
- Joint angles $\theta_t \in \mathbb{R}^{n_j}$
- (Computed) Body positions, velocities, accelerations for all tracked links

7.3 Step 2: Configure the MDP Environment

The environment configuration file (`tracking_env_cfg.py`) defines all MDP components. Key settings to verify:

Listing 2: Key environment configuration parameters.

```

1 class TrackingEnvCfg:
2     # Simulation
3     sim_dt = 0.005          # Physics timestep (200 Hz)

```

表 5: 域随机化参数。

Parameter	Sampling Distribution
<i>Physical parameters</i>	
Static friction μ_{static}	$\mathcal{U}[0.4, 1.3]$
Dynamic friction μ_{dynamic}	$\mathcal{U}[0.4, 1.1]$
Restitution e_{rest}	$\mathcal{U}[0, 0.5]$
Default joint positions (non-ankle)	$\Delta\theta_j^0 \sim \mathcal{U}[-0.01, 0.01]$ rad
Default ankle joint positions	$\Delta\theta_j^0 \sim \mathcal{U}[-0.03, 0.03]$ rad
Torso COM offset	$\Delta x \sim \mathcal{U}[-0.025, 0.025]$ m
	$\Delta y, \Delta z \sim \mathcal{U}[-0.05, 0.05]$ m
<i>Root velocity perturbations</i>	
Root linear velocity	$v_x, v_y \sim \mathcal{U}[-0.1, 0.1]$ m/s
	$v_z \sim \mathcal{U}[-0.05, 0.05]$ m/s
Push duration	$\Delta t \sim \mathcal{U}[1.0, 3.0]$ s
Root angular velocity	$\omega_x, \omega_y, \omega_z \sim \mathcal{U}[-0.1, 0.1]$ rad/s

3. BeyondMimic 代码库: https://github.com/HybridRobotics/whole_body_tracking
4. RSL-RL PPO 实现 (集成于 Isaac Lab)
5. 具备足够显存的 GPU, 以支持 16K + 个并行环境 (NVIDIA A100 或同类产品)。

7.2 步骤 1: 准备参考运动数据

你的运动匹配输出应为每个时间步的广义坐标序列。通过正向运动学将其转换为包含最大坐标 (身体姿态、速度、加速度) 的所需格式:

列表 1: 将参考运动转换为训练格式。

```

1 # Convert CSV reference to NPZ with full kinematic info
2 python scripts/csv_to_npz.py \
3     --input_file parkour_skill_climb_76cm.csv \
4     --input_fps 30 \
5     --output_name climb_76cm \
6     --headless

```

参考运动文件应包含每帧:

- 根部位移 $p_t \in \mathbb{R}^3$
- 根四元数 $r_t \in \mathbb{R}^4$
- 关节角 $\theta_t \in \mathbb{R}^{n_j}$
- (计算得到的) 所有被跟踪连杆的身体位置、速度、加速度

7.3 第二步: 配置 MDP 环境

环境配置文件 (`tracking_env_cfg.py`) 定义了所有 MDP 组件。需验证的关键设置:

列表 2: 关键环境配置参数。

```

1 class TrackingEnvCfg:
2     # Simulation
3     sim_dt = 0.005          # Physics timestep (200 Hz)

```

```

4     control_dt = 0.02      # Policy control rate (50 Hz)
5     decimation = 4        # sim_dt * decimation = control_dt
6
7     # Observations
8     obs_include_ref_joint_pos = True
9     obs_include_ref_joint_vel = True
10    obs_include_ref_pelvis_error = True
11    obs_include_proprioception = True
12    obs_include_prev_action = True
13
14    # Privileged (critic only)
15    priv_include_global_pos = True
16    priv_include_global_vel = True
17    priv_include_heightmap = True
18    heightmap_size = 0.7   # meters
19
20    # Termination
21    termination_pelvis_threshold = 0.5   # meters
22
23    # Rewards (sigma values)
24    body_pos_sigma = 0.3
25    body_ori_sigma = 0.4
26    body_linvel_sigma = 1.0
27    body_angvel_sigma = 3.14
28    anchor_pos_sigma = 0.3
29    anchor_ori_sigma = 0.4

```

```

4     control_dt = 0.02      # Policy control rate (50 Hz)
5     decimation = 4        # sim_dt * decimation = control_dt
6
7     # Observations
8     obs_include_ref_joint_pos = True
9     obs_include_ref_joint_vel = True
10    obs_include_ref_pelvis_error = True
11    obs_include_proprioception = True
12    obs_include_prev_action = True
13
14    # Privileged (critic only)
15    priv_include_global_pos = True
16    priv_include_global_vel = True
17    priv_include_heightmap = True
18    heightmap_size = 0.7   # meters
19
20    # Termination
21    termination_pelvis_threshold = 0.5   # meters
22
23    # Rewards (sigma values)
24    body_pos_sigma = 0.3
25    body_ori_sigma = 0.4
26    body_linvel_sigma = 1.0
27    body_angvel_sigma = 3.14
28    anchor_pos_sigma = 0.3
29    anchor_ori_sigma = 0.4

```

7.4 Step 3: Configure the PPO Agent

Listing 3: PPO agent configuration.

```

1 class PPOCfg:
2     # Network
3     actor_hidden_dims = [512, 256, 128]
4     critic_hidden_dims = [512, 256, 128]
5     activation = "elu"
6     init_noise_std = 1.0
7
8     # PPO
9     num_steps_per_env = 24
10    max_iterations = 20000
11    learning_rate = 1e-3
12    lr_schedule = "adaptive"
13    clip_param = 0.2
14    entropy_coef = 0.005
15    gamma = 0.99
16    lam = 0.95
17    desired_kl = 0.01
18    num_learning_epochs = 5
19    num_minibatches = 4
20    max_grad_norm = 1.0

```

7.5 Step 4: Launch Expert Training

Listing 4: Launch training for a single skill expert.

7.4 步骤 3: 配置 PPO 智能体

清单 3: PPO 智能体配置

```

1 class PPOCfg:
2     # Network
3     actor_hidden_dims = [512, 256, 128]
4     critic_hidden_dims = [512, 256, 128]
5     activation = "elu"
6     init_noise_std = 1.0
7
8     # PPO
9     num_steps_per_env = 24
10    max_iterations = 20000
11    learning_rate = 1e-3
12    lr_schedule = "adaptive"
13    clip_param = 0.2
14    entropy_coef = 0.005
15    gamma = 0.99
16    lam = 0.95
17    desired_kl = 0.01
18    num_learning_epochs = 5
19    num_minibatches = 4
20    max_grad_norm = 1.0

```

7.5 步骤 4: 启动专家训练

列表 4: 启动单一技能专家的训练。

```

1 python scripts/rs1_rl/train.py \
2   --task=Tracking-Flat-G1-v0 \
3   --registry_name org/wandb-registry-motions/climb_76cm \
4   --num_envs 16384 \
5   --headless \
6   --logger wandb \
7   --log_project_name php_experts \
8   --run_name climb_76cm_expert

```

Train one expert per skill. The full skill list used in PHP includes:

Table 6: Motion clips used in the PHP motion library.

Skill Category	Skill	Duration (s)
Locomotion	Walking / Running (0.8–3.5 m/s)	495.5
Parkour @ 1.0 m/s	Step (36 cm)	2.2
	Climb (58 cm)	12.1
	Climb (76 cm)	8.8
	Climb (94 cm)	10.3
Parkour @ 2.0 m/s	Step (36 cm)	1.6
	Climb (58 cm)	6.1
	Climb (76 cm)	4.4
	Climb (94 cm)	5.2
	Climb (125 cm)	5.9
	Dash Vault	5.0
	Speed Vault	3.1
Parkour @ 3.0 m/s	Cat Vault	1.5

7.6 Step 5: Evaluate the Expert

Listing 5: Evaluate trained expert.

```

1 python scripts/rs1_rl/play.py \
2   --task=Tracking-Flat-G1-v0 \
3   --num_envs=2 \
4   --wandb_path=org/php_experts/xxxxxxx

```

Verify that the simulated robot faithfully tracks the reference motion, paying special attention to:

- Contact events matching the reference timing
- Smooth joint trajectories without jitter
- Successful completion of the full skill clip
- Recovery from small perturbations

8 Training Pipeline: Detailed Workflow

9 From Experts to Student: The Distillation Stage

After training all skill-specific experts, the next stage (covered briefly here for context) distills them into a single multi-skill visuomotor student policy.

```

1 python scripts/rs1_rl/train.py \
2   --task=Tracking-Flat-G1-v0 \
3   --registry_name org/wandb-registry-motions/climb_76cm \
4   --num_envs 16384 \
5   --headless \
6   --logger wandb \
7   --log_project_name php_experts \
8   --run_name climb_76cm_expert

```

为每个技能训练一个专家。PHP 中使用的完整技能列表包括:

表 6: PHP 运动库中使用的运动片段。

Skill Category	Skill	Duration (s)
Locomotion	Walking / Running (0.8–3.5 m/s)	495.5
Parkour @ 1.0 m/s	Step (36 cm)	2.2
	Climb (58 cm)	12.1
	Climb (76 cm)	8.8
	Climb (94 cm)	10.3
Parkour @ 2.0 m/s	Step (36 cm)	1.6
	Climb (58 cm)	6.1
	Climb (76 cm)	4.4
	Climb (94 cm)	5.2
	Climb (125 cm)	5.9
	Dash Vault	5.0
	Speed Vault	3.1
Parkour @ 3.0 m/s	Cat Vault	1.5

7.6 步骤 5: 评估专家

清单 5: 评估训练后的专家

```

1 python scripts/rs1_rl/play.py \
2   --task=Tracking-Flat-G1-v0 \
3   --num_envs=2 \
4   --wandb_path=org/php_experts/xxxxxxx

```

确认仿真机器人准确跟踪参考运动，并特别注意以下方面：

- 与参考时序匹配的接触事件
- 关节轨迹平滑无抖动
- 成功完成完整技能片段
- 从微小扰动中恢复

8 训练流水线：详细 workflow

9 从专家到学生：蒸馏阶段

在训练完所有针对特定技能的专家后，下一阶段（此处为提供上下文而简要提及）将这些专家蒸馏为一个单一的多技能视觉运动学生策略。

Algorithm 2 Complete Expert Training Pipeline

```

1: Input: Motion-matching reference trajectories  $\{q_t^{\text{ref}}\}$ , paired terrain assets
2: Output: Trained expert policy  $\pi_i$  per skill  $i$ 
3: // Phase 1: Data Preparation
4: Retarget human motions to robot via OmniRetarget
5: Compute full kinematic features (positions, velocities, accelerations)
6: Upload to motion registry (WandB)
7: Mirror all motion clips for data augmentation
8: // Phase 2: Environment Setup
9: Configure MDP: observations, actions, rewards, terminations
10: Set domain randomization parameters (Table 5)
11: Place terrain assets with randomized geometry/pose
12: Add distractor boxes near trajectories
13: // Phase 3: Training Loop (PPO)
14: for iteration  $k = 1$  to  $K = 20,000$  do
15:   Sample start frames with adaptive sampling
16:   Reset environments to sampled reference frames
17:   for  $t = 1$  to  $T_{\text{steps}}$  do
18:     Compute observation  $o_t$  from simulator state + reference
19:     Query policy:  $a_t \sim \pi_\theta(a_t|o_t)$ 
20:     Apply PD control:  $\tau = k_p(a_t \cdot s + \theta^{\text{def}} - \theta) - k_d\dot{\theta}$ 
21:     Step simulator
22:     Compute reward  $r_t$  (Table 2)
23:     Check termination conditions
24:   end for
25:   Compute advantages via GAE
26:   Update policy and value function with PPO
27:   Update adaptive sampling statistics
28: end for

```

9.1 DAgger + RL Hybrid Distillation

The student policy observes:

- Proprioception (pelvis gravity vector, angular velocity, joint states, previous action)
- Depth images (rendered via NVIDIA Warp at 58×87 resolution)
- 2D velocity commands

The distillation loss combines DAgger (behavior cloning from expert actions) with PPO:

$$L = \lambda_{\text{PPO}} L_{\text{PPO}} + \lambda_D L_D, \quad \lambda_{\text{PPO}} + \lambda_D = 1 \quad (6)$$

A warmup curriculum linearly decreases λ_D during the first half of training:

$$\lambda_D(k) = \max\left(0.1, 1 - \frac{k}{K/2}\right) \quad (7)$$

9.2 Why DAgger Alone Is Insufficient

For highly dynamic skills like climbing and vaulting, DAgger fails because:

- Per-step imitation loss does not account for episode outcomes.

Algorithm 2 Complete Expert Training Pipeline

```

1: Input: Motion-matching reference trajectories  $\{q_t^{\text{ref}}\}$ , paired terrain assets
2: Output: Trained expert policy  $\pi_i$  per skill  $i$ 
3: // Phase 1: Data Preparation
4: Retarget human motions to robot via OmniRetarget
5: Compute full kinematic features (positions, velocities, accelerations)
6: Upload to motion registry (WandB)
7: Mirror all motion clips for data augmentation
8: // Phase 2: Environment Setup
9: Configure MDP: observations, actions, rewards, terminations
10: Set domain randomization parameters (Table 5)
11: Place terrain assets with randomized geometry/pose
12: Add distractor boxes near trajectories
13: // Phase 3: Training Loop (PPO)
14: for iteration  $k = 1$  to  $K = 20,000$  do
15:   Sample start frames with adaptive sampling
16:   Reset environments to sampled reference frames
17:   for  $t = 1$  to  $T_{\text{steps}}$  do
18:     Compute observation  $o_t$  from simulator state + reference
19:     Query policy:  $a_t \sim \pi_\theta(a_t|o_t)$ 
20:     Apply PD control:  $\tau = k_p(a_t \cdot s + \theta^{\text{def}} - \theta) - k_d\dot{\theta}$ 
21:     Step simulator
22:     Compute reward  $r_t$  (Table 2)
23:     Check termination conditions
24:   end for
25:   Compute advantages via GAE
26:   Update policy and value function with PPO
27:   Update adaptive sampling statistics
28: end for

```

9.1 DAgger + RL 混合蒸馏

学生策略观测:

- 本体感觉（骨盆重力向量、角速度、关节状态、上一动作）

深度图像（通过 NVIDIA Warp 渲染，分辨率为 58 × 87）

- 二维速度指令

蒸馏损失结合了 DAgger（来自专家动作的行为克隆）与 PPO:

$$L = \lambda_{\text{PPO}} L_{\text{PPO}} + \lambda_D L_D, \quad \lambda_{\text{PPO}} + \lambda_D = 1 \quad (6)$$

一种热身策略在训练的前半段将 λ_D 线性降低:

$$\lambda_D(k) = \max\left(0.1, 1 - \frac{k}{K/2}\right) \quad (7)$$

9.2 为何仅靠 DAgger 不足

对于攀爬和跨越等高度动态的技能，DAgger 失败的原因是:

逐步模仿损失并未考虑回合结果。

- Critical high-torque bursts (spanning only a few timesteps) receive insufficient gradient signal.
- Actions symmetrically above/below the reference receive identical loss, even though only one direction succeeds.

PPO provides an episode-level success signal that encourages the student to exploit expert behaviors at critical moments.

9.3 Student Architecture

Table 7: Student policy architecture.

Component	Specification
Depth backbone	3-layer CNN + Global Average Pooling
Depth output dim	32
Depth input resolution	58×87
Student MLP hidden dims	[2048, 1024, 512, 256, 128]
Critic MLP hidden dims	[512, 256, 128]
Activation	ELU
Initial noise std	0.01

10 Troubleshooting and Best Practices

10.1 Common Issues

1. Expert fails to converge on difficult skills.

- Ensure adaptive sampling is enabled.
- Verify the reference motion is physically feasible for the robot.
- Check that terrain assets are correctly placed relative to the reference.

2. Jittery behavior in simulation.

- Reduce the action scale if torques are saturating.
- Increase action smoothness penalty weight.
- Verify PD gains are appropriate for the robot’s actuator model.

3. Large sim-to-real gap.

- Avoid excessive domain randomization.
- Calibrate PD gains against hardware.
- Ensure the simulation timestep is small enough (200 Hz recommended).

4. Student fails on skills the expert succeeds at.

- Add RL to distillation (do not rely on DAgger alone).
- Relax termination thresholds for mirrored executions.
- Keep the DAgger loss active throughout training for regularization.

- 关键的高转矩爆发（仅持续数个时间步）接收到的梯度信号不足。
- 即使仅有一个方向成功，参考值上方和下方对称的动作也会获得相同的损失。

PPO 提供回合级成功信号，激励学生模型在关键时刻利用专家行为。

9.3 学生架构

表 7：学生策略架构

Component	Specification
Depth backbone	3-layer CNN + Global Average Pooling
Depth output dim	32
Depth input resolution	58×87
Student MLP hidden dims	[2048, 1024, 512, 256, 128]
Critic MLP hidden dims	[512, 256, 128]
Activation	ELU
Initial noise std	0.01

10 故障排查与最佳实践

10.1 常见问题

1. 专家在困难技能上无法收敛。

- 确保自适应采样已启用。
- 验证参考运动在物理上对机器人可行。
- 检查地形资产是否相对于参考位置正确放置。

2. 仿真中的抖动行为

- 若力矩出现饱和，则减小动作尺度。
- 增加动作平滑惩罚权重。
- 验证 PD 增益是否适用于机器人的执行器模型。

3. 巨大的仿真-真实差距。

- 避免过度域随机化。
- 对照硬件校准 PD 增益。
- 确保仿真时间步长足够小（推荐 200 Hz）。

4. 学生在专家擅长的技能上失败。

- 向蒸馏中加入强化学习（不要仅依赖 DAgger）。
- 放宽镜像执行的终止阈值。

保持 DAgger 损失在整个训练过程中激活作为正则化。

10.2 Key Takeaways

Summary of Best Practices

- Use a unified MDP formulation—no per-skill reward tuning.
- Enable adaptive sampling for contact-rich skills.
- Set action scale to 1 for maximum expressiveness.
- Use privileged global position/velocity in the expert critic.
- Apply lightweight domain randomization to preserve motion quality.
- Train 20K iterations with 16K+ parallel environments.
- Mirror all motion clips for data augmentation.
- During distillation, never drop the DAgger loss entirely.

11 References

- [1] Z. Wu, X. Huang, L. Yang, et al., “Perceptive Humanoid Parkour: Chaining Dynamic Human Skills via Motion Matching,” 2025. <https://php-parkour.github.io/>
- [2] Q. Liao, T. E. Truong, X. Huang, et al., “BeyondMimic: From Motion Tracking to Versatile Humanoid Control via Guided Diffusion,” arXiv:2508.08241, 2025. <https://beyondmimic.github.io/>
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- [4] L. Yang, X. Huang, Z. Wu, et al., “OmniRetarget: Interaction-Preserving Data Generation for Humanoid Whole-Body Loco-Manipulation,” arXiv:2509.26633, 2025.
- [5] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, “Proximal Policy Optimization Algorithms,” arXiv:1707.06347, 2017.
- [6] BeyondMimic Code Repository: https://github.com/HybridRobotics/whole_body_tracking

10.2 关键点

最佳实践总结

- 使用统一的 MDP 形式化——无需为每个技能分别调整奖励函数。
- 启用接触丰富技能的适应性采样。
- 将动作缩放设为 1 以获得最大表现力。
- 在专家评论家中使用特权全局位置/速度。
- 应用轻量级域随机化以保持运动质量。
- 使用 16K + 个并行环境训练 20K 次迭代。
- 对所有运动片段进行镜像以进行数据增强。
- 在蒸馏过程中，切勿完全丢弃 DAgger 损失。

11 参考文献

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